A Fourth-Moment Eigenvalue Bound for Balanced Bi-Independent Pairs

Abstract

Let $G = (V_1 \cup V_2, E)$ be an r-regular bipartite graph with $|V_1| = |V_2| = n$, and let $r = \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ be the singular values of its $n \times n$ bipartite adjacency matrix. The open-problems formulation associated with Laurent, Polak and Vargas asks for a closed-form eigenvalue upper bound on the balanced bi-independence number $\alpha_{\mathrm{bal}}(G)$ which is valid for all regular bipartite graphs and can be strictly smaller than $4\mathrm{b}h(G) = 2n\sigma_2/(r + \sigma_2)$.

This note proves such a bound. If $d = n - r$ and

$$\Gamma_4(G) = \frac{d^4 + \sum_{i=2}^{n} \sigma_i^4 - nd(2d-1)}{4},$$

then, for $r > 0$,

$$\alpha_{\mathrm{bal}}(G) \leq \min\left\{\frac{2n\sigma_2}{r + \sigma_2}, 1 + \sqrt{1 + 8\sqrt{\Gamma_4(G)}}\right\}.$$

The quantity $\Gamma_4(G)$ is exactly the number of unoriented 4-cycles in the bipartite complement of G, and is a closed-form expression in the adjacency spectrum. The bound is strictly better than $4\mathrm{b}h(G)$ on the bipartite complements of projective-plane incidence graphs.

Status: this is a solution to the open-problems formulation as just stated. If one instead requires a formula depending only on n, r, σ_2, or a formula arising from the same symmetric SDP templates studied by Laurent–Polak–Vargas, then this should be read as partial progress rather than a complete solution.

1 Introduction and status of the result

Let $G = (V_1 \cup V_2, E)$ be a bipartite graph. A pair (A, B), with $A \subseteq V_1$ and $B \subseteq V_2$, is called a *bi-independent pair* if there is no edge of G between A and B. It is *balanced* if $|A| = |B|$. We write

$$\alpha_{\mathrm{bal}}(G) = \max\{|A| + |B| : (A, B) \text{ is a balanced bi-independent pair in } G\}.$$

Equivalently, $\alpha_{\mathrm{bal}}(G) = 2a(G)$, where $a(G)$ is the maximum integer a for which there is a bi-independent pair (A, B) with $|A| = |B| = a$.

For an r-regular bipartite graph on two sides of size n, the standard second-singular-value argument gives

$$\alpha_{\mathrm{bal}}(G) \leq 4\mathrm{b}h(G) = \frac{2n\sigma_2}{r + \sigma_2}. \tag{1}$$

The question addressed here is whether one can obtain a closed-form spectral upper bound for $\alpha_{\mathrm{bal}}(G)$ that remains valid for every regular bipartite graph and is sometimes strictly smaller than (1).

The theorem below gives such a formula. It is not a σ_2-only bound: it uses the full fourth spectral moment, equivalently the number of 4-cycles in the bipartite complement. Thus it solves the stated open-problems formulation, while representing only partial progress for the stronger variant in which the improved bound is required to depend solely on n, r, σ_2 or to arise from the same SDP symmetrizations.

2 Notation and statement

Throughout the paper, $G = (V_1 \cup V_2, E)$ is an r-regular bipartite graph with $|V_1| = |V_2| = n$. Let $M \in \{0,1\}^{n\times n}$ be its bipartite adjacency matrix, rows indexed by V_1 and columns by V_2. Thus

$$M\mathbf{1} = M^T\mathbf{1} = r\mathbf{1}.$$

Let

$$r = \sigma_1(M) \geq \sigma_2(M) \geq \cdots \geq \sigma_n(M) \geq 0$$

be the singular values of M; when there is no risk of ambiguity we write σ_i for $\sigma_i(M)$. Let

$$d = n - r, \qquad N = J - M,$$

where J is the $n \times n$ all-ones matrix. Then N is the bipartite adjacency matrix of the bipartite complement $\overline{G}^b$ of G, and $\overline{G}^b$ is d-regular.

For a bipartite graph H, write $C_4(H)$ for its number of unoriented simple 4-cycles. Define

$$\Gamma_4(G) := \frac{d^4 + \sum_{i=2}^n \sigma_i^4 - nd(2d-1)}{4}. \tag{2}$$

Theorem 1 (Fourth-moment balanced bound)**.** *Let $G = (V_1 \cup V_2, E)$ be an r-regular bipartite graph with $|V_1| = |V_2| = n$. If $r > 0$, then*

$$\alpha_{\mathrm{bal}}(G) \leq B_4(G) := \min\left\{\frac{2n\sigma_2}{r+\sigma_2}, 1 + \sqrt{1 + 8\sqrt{\Gamma_4(G)}}\right\}. \tag{3}$$

If $r = 0$, then G is the empty bipartite graph, $\alpha_{\mathrm{bal}}(G) = 2n$, and the second term in (3) gives the exact value $2n$.

Since $\alpha_{\mathrm{bal}}(G)$ is an even integer, the integer-valued refinement

$$\alpha_{\mathrm{bal}}(G) \leq 2\left\lfloor \min\left\{\frac{n\sigma_2}{r+\sigma_2}, \frac{1 + \sqrt{1 + 8\sqrt{\Gamma_4(G)}}}{2}\right\}\right\rfloor \tag{4}$$

is also valid when $r > 0$.

3 Proof of the bound

We split the proof into three elementary pieces: the old second-singular-value estimate, the spectral formula for $C_4(\overline{G}^b)$, and the observation that a balanced bi-independent pair forces a square biclique in $\overline{G}^b$.

Lemma 2 (Second-singular-value estimate)**.** *Assume $r > 0$. Let (A, B) be a balanced bi-independent pair in G with $|A| = |B| = a$. Then*

$$a \leq \frac{n\sigma_2}{r+\sigma_2}.$$

Consequently,

$$\alpha_{\mathrm{bal}}(G) \leq \frac{2n\sigma_2}{r+\sigma_2}.$$

Proof. If $a = 0$, the claim is immediate. Assume $a > 0$. Let

$$x = \chi_A - \frac{a}{n}\mathbf{1}, \qquad y = \chi_B - \frac{a}{n}\mathbf{1}.$$

Then $x \perp \mathbf{1}$, $y \perp \mathbf{1}$, and
$$\|x\|^2 = \|y\|^2 = a\left(1 - \frac{a}{n}\right).$$
Since there are no edges of G between A and B,
$$0 = \chi_A^T M \chi_B.$$
Using $M\mathbf{1} = M^T\mathbf{1} = r\mathbf{1}$ and expanding around the all-ones direction gives
$$0 = \frac{ra^2}{n} + x^T M y.$$
On the subspace orthogonal to $\mathbf{1}$, the operator norm of M is σ_2. Hence
$$\frac{ra^2}{n} = |x^T M y| \le \sigma_2 \|x\|\|y\| = \sigma_2 a\left(1 - \frac{a}{n}\right).$$
Dividing by $a > 0$ and rearranging gives
$$\frac{ra}{n} \le \sigma_2\left(1 - \frac{a}{n}\right), \qquad \text{hence} \qquad a \le \frac{n\sigma_2}{r + \sigma_2}.$$

□

Lemma 3 (Spectrum and 4-cycles in the bipartite complement)**.** *The quantity $\Gamma_4(G)$ in (2) is exactly $C_4(\overline{G}^b)$, the number of unoriented 4-cycles in the bipartite complement of G.*

Proof. The complement matrix $N = J - M$ satisfies
$$N\mathbf{1} = N^T\mathbf{1} = d\mathbf{1}.$$
Moreover, if $z \perp \mathbf{1}$, then $Jz = 0$, and hence $Nz = -Mz$. The subspace $\mathbf{1}^\perp$ is invariant under M and M^T, because $M\mathbf{1} = M^T\mathbf{1} = r\mathbf{1}$. Therefore the singular values of N are
$$d, \sigma_2, \ldots, \sigma_n,$$
not necessarily listed in decreasing order. It follows that
$$\operatorname{tr}\big((NN^T)^2\big) = d^4 + \sum_{i=2}^{n} \sigma_i^4. \tag{5}$$

For $u, v \in V_1$, let
$$c_{uv} = |N_{\overline{G}^b}(u) \cap N_{\overline{G}^b}(v)|$$
be the number of common neighbors of u and v in $\overline{G}^b$. Then $c_{uv} = (NN^T)_{uv}$ and $c_{uu} = d$. Hence
$$\sum_{u,v \in V_1} c_{uv}^2 = \operatorname{tr}\big((NN^T)^2\big),$$
and therefore
$$\sum_{\{u,v\} \subseteq V_1} c_{uv}^2 = \frac{\operatorname{tr}((NN^T)^2) - nd^2}{2}. \tag{6}$$
Also,
$$\sum_{\{u,v\} \subseteq V_1} c_{uv} = n\binom{d}{2}, \tag{7}$$
because each right-side vertex of $\overline{G}^b$ has d left-side neighbors and contributes one to c_{uv} for each unordered pair among those d neighbors.

Every unoriented 4-cycle in a bipartite graph is uniquely determined by its two left vertices and its two right vertices. Thus

$$C_4(\overline{G}^b) = \sum_{\{u,v\}\subseteq V_1} \binom{c_{uv}}{2}.$$

Using (6) and (7), we obtain

$$\begin{aligned} C_4(\overline{G}^b) &= \frac{1}{2} \sum_{\{u,v\}\subseteq V_1} (c_{uv}^2 - c_{uv}) \\ &= \frac{1}{2}\left(\frac{\operatorname{tr}((NN^T)^2) - nd^2}{2} - n\binom{d}{2}\right) \\ &= \frac{\operatorname{tr}((NN^T)^2) - nd(2d-1)}{4}. \end{aligned}$$

Substituting (5) gives exactly (2). □

Lemma 4 (A balanced pair forces 4-cycles)**.** *If G contains a balanced bi-independent pair (A, B) with $|A| = |B| = a$, then*

$$\binom{a}{2}^2 \le \Gamma_4(G).$$

Consequently,

$$a \le \frac{1 + \sqrt{1 + 8\sqrt{\Gamma_4(G)}}}{2}.$$

Proof. The condition $A \times B \cap E(G) = \emptyset$ says precisely that every edge between A and B is present in the bipartite complement $\overline{G}^b$. Hence $\overline{G}^b$ contains a copy of $K_{a,a}$ with left part A and right part B. Every choice of two vertices from A and two vertices from B gives a distinct unoriented 4-cycle in this $K_{a,a}$. Therefore

$$\binom{a}{2}^2 \le C_4(\overline{G}^b) = \Gamma_4(G),$$

where the equality is Lemma 3. Taking square roots gives

$$\frac{a(a-1)}{2} \le \sqrt{\Gamma_4(G)}.$$

Solving this quadratic inequality in a gives the stated bound. □

Proof of Theorem 1. Let (A, B) be an optimal balanced bi-independent pair and write $|A| = |B| = a$, so $\alpha_{\text{bal}}(G) = 2a$. For $r > 0$, Lemma 2 gives

$$2a \le \frac{2n\sigma_2}{r + \sigma_2},$$

and Lemma 4 gives

$$2a \le 1 + \sqrt{1 + 8\sqrt{\Gamma_4(G)}}.$$

Taking the minimum of these two valid upper bounds proves (3). Since a is an integer, applying the floor to the corresponding upper bound on a proves (4).

If $r = 0$, then G has no edges, so $\alpha_{\text{bal}}(G) = 2n$. In this case $d = n$, the bipartite complement is $K_{n,n}$, and

$$\Gamma_4(G) = \binom{n}{2}^2.$$

Therefore

$$1+\sqrt{1+8\sqrt{\Gamma_4(G)}} = 1+\sqrt{1+8\binom{n}{2}} = 1+\sqrt{(2n-1)^2} = 2n,$$

which proves the asserted exactness in the empty case. □

4 The bound is an eigenvalue bound

The displayed formula for $\Gamma_4(G)$ uses singular values of the bipartite adjacency block. Equivalently, it may be written directly in terms of the eigenvalues of the full adjacency matrix

$$A_G = \begin{pmatrix} 0 & M \\ M^T & 0 \end{pmatrix}.$$

The eigenvalues of A_G are

$$\pm r, \quad \pm\sigma_2, \ldots, \pm\sigma_n,$$

with multiplicities. Hence

$$\sum_{i=2}^{n} \sigma_i^4 = \frac{1}{2}\operatorname{tr}(A_G^4) - r^4.$$

Thus

$$\Gamma_4(G) = \frac{d^4 + \frac{1}{2}\operatorname{tr}(A_G^4) - r^4 - nd(2d-1)}{4}. \tag{8}$$

This shows explicitly that $B_4(G)$ is a closed-form expression in the adjacency spectrum of G.

5 Strict improvement on an infinite family

Proposition 5 (Bipartite complements of projective-plane incidence graphs)**.** *Let $q \geq 2$ be a prime power, and let H_q be the point-line incidence bipartite graph of a projective plane of order q. Thus each side has*

$$n = q^2 + q + 1$$

vertices, and H_q is $(q+1)$-regular. Let G_q be the bipartite complement of H_q. Then

$$\alpha_{\mathrm{bal}}(G_q) = B_4(G_q) = 2,$$

whereas

$$4\mathrm{b}h(G_q) = \frac{2(q^2+q+1)\sqrt{q}}{q^2+\sqrt{q}} > 2.$$

Consequently, the bound in Theorem 1 is strictly stronger than $4\mathrm{b}h$ on an infinite family of regular bipartite graphs.

Proof. The incidence graph H_q has no 4-cycles: two distinct points of a projective plane determine a unique line, so two distinct point-vertices cannot have two common line-neighbors. Since G_q is the bipartite complement of H_q, the complement $\overline{G_q}^{\,b}$ is exactly H_q. Therefore

$$\Gamma_4(G_q) = C_4(H_q) = 0.$$

The fourth-moment term in Theorem 1 is then

$$1+\sqrt{1+8\sqrt{0}} = 2,$$

so $B_4(G_q) \le 2$. On the other hand, G_q is not complete bipartite, because H_q has edges. Thus G_q has at least one nonedge, which gives a balanced bi-independent pair with one vertex on each side. Hence $\alpha_{\mathrm{bal}}(G_q) \ge 2$. Combining the two inequalities gives $\alpha_{\mathrm{bal}}(G_q) = B_4(G_q) = 2$.

Let P_q be the incidence matrix of the projective plane. Since every point is incident with $q+1$ lines and every pair of distinct points lies on a unique common line,

$$P_q P_q^T = qI + J.$$

Therefore the singular values of P_q are $q+1$ once and $\sqrt{q}$ with multiplicity $n-1$. The graph G_q has degree

$$r = n - (q+1) = q^2,$$

and its nontrivial singular values are again $\sqrt{q}$, because passing to the bipartite complement preserves the nontrivial singular values. Hence

$$4\mathrm{b}h(G_q) = \frac{2n\sqrt{q}}{r+\sqrt{q}} = \frac{2(q^2+q+1)\sqrt{q}}{q^2+\sqrt{q}}.$$

This is greater than 2 exactly when

$$(q^2+q+1)\sqrt{q} > q^2+\sqrt{q},$$

or equivalently

$$q\big((q+1)\sqrt{q} - q\big) > 0,$$

which holds for all $q \ge 2$. □

6 Discussion and limitations

The proof uses one new observation: balancedness turns a bi-independent pair in G into a square biclique in the bipartite complement. A $K_{a,a}$ contains $\binom{a}{2}^2$ four-cycles, and in a regular bipartite graph the total number of four-cycles is determined by the fourth spectral moment. This is why the estimate is sensitive to a feature that the second-singular-value quotient alone may miss.

The price is that the theorem uses the full fourth spectral moment, or equivalently all nontrivial singular values through $\sum_{i=2}^n \sigma_i^4$. Therefore it should not be advertised as a σ_2-only strengthening of the Laurent–Polak–Vargas expression. It also does not claim to improve the SDP relaxations themselves. Rather, it provides an additional closed-form spectral obstruction, and taking the minimum with the existing expression gives a universal spectral bound that is never worse than $4\mathrm{b}h(G)$ and is sometimes strictly better.

This is the sense in which the note obtains a solution to the extracted open problem. Under the narrower interpretation "find a better formula using only n, r, σ_2" or "find a better symmetric SDP specialization of the same type", it should instead be viewed as partial progress.

References

[1] M. Laurent, S. Polak, and L. F. Vargas, *Semidefinite Approximations for Bicliques and Bi-Independent Pairs*, Mathematics of Operations Research **50**(1), 537–572, 2025. doi:10.1287/moor.2023.0046.

[2] Open Problems in Operations Research, Problem 114, "Eigenvalue upper bound for balanced independence number." `https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4392748341_p0&page=10&n=114`.